Band-1 Hadamard Order 6196 — Miyamoto (Paley-II H_1548 + C_1550 over GF(1549)); Miyamoto-only Bonus (NOT in SPK); Gap 2052 Still OPEN

H_1548/C_1550/H_6196 Miyamoto PASS; Paley-I/Double SKIP; Paley-I/II on 6196 SKIP; 6196 Miyamoto-only (NOT in SPK); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-15

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_5956_Miyamoto`; `2026-09-15_Band1_Hadamard_5924_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_6196_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 1548 and 6196 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with preferred Paley-II seed H_{1548} = Paley-II($q = 773$) and symmetric Paley conference C_{1550} over **GF(**1549**)** (prime field, $q = 1549 \equiv 1 \pmod 4$). Paley-I for the seed ($q = 1547 = 7 \cdot 13 \cdot 17$) **FAIL**s; Double half=774 **FAIL**s ($q = 773 \equiv 1 \pmod 4$). Classical SPK paths for order 6196 **FAIL**: Paley-I ($q = 6195 = 3 \cdot 5 \cdot 7 \cdot 59$ not prime); Paley-II ($q = 3097 = 19 \cdot 163$ not prime); Kronecker (4×1549). Therefore order **6196 is a Miyamoto-only bonus (NOT in SPK)** (same honesty pattern as 5956/5812/5476). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. `clay_closed = false`.

Abstract

Finite constructions of orders **1548** and **6196** via Miyamoto.

- H_{1548} = Paley-II ($q = 773$) — preferred; Paley-I / Double SKIP
- C_{1550} = Paley conference over GF(1549) prime field
- H_{6196} = Miyamoto(H_{1548}, C_{1550})
- Paley-I / Paley-II for 6196: **SKIP**

SPK: 6196 Miyamoto-only bonus (NOT in SPK). Seed 1548 enters SPK via Paley-II. Prior seed 1488 (from 5956 ship) retained in ONBOX.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_6196.py`

Results. overall_pass=true; units 74/74; wall 194.5835s; all max|HH^T-nI|=0.

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Arithmetic

- Target $n = 6196 = 4 \cdot 1549$; 1549 prime $\equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{1548} and C_{1550} (Paley conference over prime field 1549).
- Seed preferred: Paley-II $q = 773$ (prime $\equiv 1 \pmod 4$); $2(q+1) = 1548$.
- Paley-I seed $q = 1547$ **FAIL** ($7 \cdot 13 \cdot 17$); Double half=774 **FAIL**.
- SPK: Paley-I $q = 6195$ **FAIL**; Paley-II $q = 3097$ **FAIL**; Kronecker **FAIL** → **Miyamoto-only**.

Constructions (verified)

Object	Method	max|HH^T-nI|	Wall (s)
H_1548 Paley-II	Paley-II q=773 (preferred)	0	2.0113
H_1548 Paley-I	SKIP q=1547=$7 \cdot 13 \cdot 17$ not prime	—	—
H_1548 Double	SKIP q=773 $\equiv$ 1 mod 4	—	—
C_1550	Paley conf GF(1549) q=1549	0	2.7591
H_6196 Miyamoto	Miyamoto(H_1548, C_1550)	0	187.3521
H_6196 Paley-I	SKIP q=6195=$3 \cdot 5 \cdot 7 \cdot 59$ not prime	—	—
H_6196 Paley-II	SKIP q=3097=$19 \cdot 163$ not prime	—	—

SPK closure note

Seeds include prior SPK ships through **5956** lineage (**1488**/744 from that ship; **NOT** 5956) + verified **1548** (Paley-II); **6196 NOT folded** (Miyamoto-only); **exclude** Miyamoto-only bonuses in-

cluding **5956/5812/5476/5284/6196**. Known count 1297. Next constructive gap **2052** (OPEN).

Quantity	Value
6196 in SPK	**False** (Miyamoto-only bonus)
1548 in SPK	**True** (Paley-II)
1488 in SPK	**True** (prior Paley-I)
5956 in SPK	**False** (prior Miyamoto-only)
5924 in SPK	**True** (prior Paley-I)
Next constructive gap	**2052** (OPEN)

Next Miyamoto targets (after q=1549)

q	n=4q	need H	build notes
1553	6212	1552	[‘Kronecker(4x388)’]
1597	6388	1596	[‘PaleyII(q=797)’]
1601	6404	1600	[‘Kronecker(4x400)’]
1609	6436	1608	[‘PaleyI(q=1607)’]
1613	6452	1612	[]
1621	6484	1620	[‘PaleyI(q=1619)’, ‘PaleyII(q=809)’]

Honesty

clay_closed = false · GATE-04 LOCKED · Hadamard conjecture OPEN · gap 2052 OPEN · no RH/P!=NP/YM/Frankl/NS/AC/Clay-closed claims.